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Stock price prediction based on relative strength index, moving average convergence divergence and XGBoost model

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Abstract: Stock price prediction is crucial in shaping investment strategies, facilitating risk management, and aiding decision-making processes for both investors and financial institutions. Based on the trading data of ZTE's stock from 2015 to 2022, this paper utilizes XGBoost model to forecast the stock price and compares it with the actual value. The research includes calculation of exponential moving average and simple moving average, analysis of relative strength index, calculation of moving average convergence divergence, and application of XGBoost regression model for stock price prediction. By comparing the predicted value with the actual value, the prediction ability of XGBoost model is verified. The results of this study provide a feasible stock price prediction method for investors and financial institutions, and have important reference value for formulating investment strategies and managing risks.

Key Words: Stock price prediction, exponential moving average, simple moving average, relative strength index, moving average convergence divergence, XGBoost

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1 Introduction

Stock price prediction is of great significance to investors, financial institutions and government regulators. Investors want higher returns through accurate forecasts, financial institutions need to manage risks effectively, and government regulators want to maintain market stability by monitoring market trends. However, the volatility of the stock market is affected by a variety of factors, including economic indicators, corporate performance, political events, and market sentiment, and its complexity and randomness make it an extremely challenging problem to accurately predict stock prices.

ZTE is one of the representative enterprises in China's communication equipment industry, which is one of the key industries in China's current development and the commanding heights of today's industrial development. In recent years, China has introduced a lot of policies to promote the development of the industry, and after decades of accumulation, China's communication equipment industry has also entered a stage of rapid development. Due to China's weak strength in some scientific and technological fields, communication equipment enterprises are often subject to many restrictions in the development process. In this context, ZTE's cash flow and net profit are unstable, and the stock price fluctuates greatly. Some investors gain a lot while others suffer heavy losses, which makes ZTE's stock forecast very important [1-3].

For the prediction of stock price, relative strength index (RSI) and moving average convergence divergence (MACD) have always been important indicators of stock price prediction. The principle of RSI is to judge the relative strength of the buyer and seller in the financial market through the rise and fall of the stock closing price or index over a period of time, and to predict the stock price or future trend change [4], [5]. When the RSI value is between 20 and 80, it is considered that the forces are relatively balanced, and no operation is performed at this time. When the RSI value is lower than 20, it is considered that the market sentiment is relatively weak, the seller's power is dominant, and the market may reverse. At this time, the market should be bought against the market. On the contrary, when the RSI value is greater than 80, it is considered that the market sentiment is relatively strong, the buyer's power is dominant, and the market may reverse. At this time, the market should be sold against the market operation. The MACD index essentially reflects the adhesion and separation between the short-term average price line and the long-term average price line of the stock price, reflecting the trend and intensity of the stock trend. The MACD analysis method uses more indicators, which has the advantages of fewer error signals and higher final returns. At the same time, it is more realistic for MACD indicators to assign higher weights to the prices that are closer to them [6].

In addition, Estimation of distribution algorithm (EDA) is a new research direction of evolutionary algorithm [7], [8]. EDA algorithm replaces traditional genetic operations such as crossover and variation through learning and sampling of probabilistic models, and uses statistical learning methods to build a probabilistic model from the macro perspective of a population to describe the distribution of candidate solutions in the solution space. Then, under the obtained probabilistic model, new populations are generated through random sampling, and the evolution of populations is realized through repeated iterations. Constantly approaching the optimal solution of the problem. Extreme Gradient Boosting

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(XGBoost) algorithm was put forward by Dr Chen Tianqi at the university of Washington, the algorithm by Gradient Boosting regression tree (GBRT) algorithm evolved [9], [10]. XGBoost algorithm is the realization of GBRT idea, the biggest feature is that it can accelerate the calculation through parallel learning, and improve the objective function to improve the prediction accuracy. Neural networks are becoming more and more popular in data prediction, and XGBoost algorithm can obtain better prediction results under the condition of limited training samples and training time. Compared to deep neural networks, XGBoost can handle tabular data better and has stronger interpretability. A large number of experiments show that this model can improve the prediction accuracy while using the least resources to solve the problem.

Taking the closing price of ZTE Corporation as an example, this paper used RSI and MACD indicators, combined with Exponential Moving Average (EMA) algorithm and XGBoost algorithm to analyze and predict the stock price.

2 Theory and Hypothesis

2.1 Exponential Moving Average (EMA) and Simple Moving Average (SMA)

Moving average is an important index widely used in stock price analysis. It helps to identify the trend and strength of price movement by smoothing the price data. In stock market research, EMA is a widely used technical indicator that is mainly used to smooth out changes in stock closing prices and eliminate noise and irrelevant signals. SMA smooths historical data and then uses the smoothed values as future predictions, which has the advantage of being simple and fast to calculate. Compared to the SMA, the EMA puts more emphasis on the latest closing price data and gives it a greater weight to reflect the market's volatility more quickly. The daily moving average is calculated based on the average of the closing prices in the past period of time. It is a statistical average of the closing prices in the past day, which effectively smoothen the price fluctuations and helps us to observe the long-term trend of the price more accurately. The EMA is calculated as

$$EMA_n = \alpha * Close + (1-\alpha) * EMA_{n-1}, \tag{1}$$

where n represents the number of days of the moving average, α is the smoothness index, its value can be calculated by the following formula: $\alpha = 2/n+1$. $Close$ represents the current closing price, EMA_{n-1} indicates the EMA value of the previous period. The SMA is calculated by:

$$SMA_n = (Close_1 + Close_2 + \cdots + Close_n)/n, \tag{2}$$

where n represents the number of days of the moving average, $Close_1, Close_2, \cdots, Close_n$ indicates the closing price for the past day.

By calculating the 9-day exponential moving average EMA and the 5-day, 10-day, 15-day and 30-day SMA, this paper can more comprehensively understand the trend and strength of prices, and provide a quantitative indicator basis. For decision makers to develop investment strategy and risk control. In particular, the characteristics of the exponential moving average make it more sensitive to changes in the market, so it plays an important role in market analysis and trading decisions. The application of the daily moving average is mainly focused on capturing the long-term trend of price changes and providing a reference to the long-term trend of the market.

2.2 Relative strength index (RSI)

The RSI Index is widely used in the stock market to measure buying and selling power and to determine whether stocks are over-bought or over-sold. In stock forecasting and technical analysis, RSI indicators can be used to predict the trend of a stock. When the RSI indicator is in overbought territory (usually above 80), it indicates that the stock may have been overbought and there is a potential opportunity for a correction or correction in the market; Conversely, when the RSI indicator is in oversold territory (usually below 20), it means that the stock may have been oversold and the market may have an opportunity for a rally or recovery. Second, RSI indicators can be used to determine the timing of buying and selling stocks. By observing the intersections and changing trends of the RSI lines, investors can use them as a reference for buying and selling stocks. When the RSI line breaks the oversold area from below up, it may be a buy signal; Conversely, when the RSI line breaks the overbought area from the top down, it may be a sell signal. To calculate the RSI, first calculate the movement of the stock price on each trading day δ:

$$\delta = Close_n - Close_{n-1}. \tag{3}$$

Divide change δ into rising part δ_{gain} and falling part δ_{loss}:

$$\begin{aligned} \delta_{gain} &= \delta(if\ \delta > 0)\ or\ \delta_{gain} = 0(if\ \delta \le 0), \\ \delta_{loss} &= -\delta(if\ \delta < 0)\ or\ \delta_{loss} = 0(if\ \delta \ge 0). \end{aligned} \tag{4}$$

Calculate the average of the rise and fall, usually using the EMA:

$$\begin{aligned} average_\delta_{gain} &= EMA(\delta_{gain}, n), \\ average_\delta_{loss} &= EMA(\delta_{loss}, n). \end{aligned} \tag{5}$$

Calculate RSI:

$$\begin{aligned} RS &= average_\delta_{gain} / average_\delta_{loss}, \\ RSI &= 100 - (100/(1+RS)). \end{aligned} \tag{6}$$

RS is the ratio of rising and falling averages, and RSI is the percentage form that converts RS into an indicator. Based on the value of the RSI and the judgment of the overbought and oversold areas, some trading signals and strategies can be generated. For example, when the RSI rises from an oversold area to an overbought area, a buy signal may be generated; When the RSI falls from an overbought area to an oversold area, a sell signal may be generated. By observing changes in

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the RSI on a chart, you can aid in making trading decisions and managing risk.

2.3 Moving average convergence divergence (MACD)

The MACD is one of the technical indicators commonly used by investors to predict the trend of the stock market and help investors make decisions when buying or selling stocks. The MACD uses convergence and separation of short- and long-term EMA for its analysis. The MACD consists of three main components: the MACD line, the signal line, and the MACD histogram. Among them, the MACD line is usually obtained by calculating the difference between the short-term (generally 12 days) EMA and the long-term (generally 26 days) EMA, also known as the DIF; The signal line is usually obtained by calculating the nine-day moving average (i.e. DEA); The MACD histogram is calculated by subtracting the signal line from twice the MACD line, and is usually expressed in OSC. The above indicators can be obtained from the following formula:

$$\begin{aligned} MACD = DIF = EMA_{12} - EMA_{26}, \\ DEA = EMA(DIF, signal_period), \\ OSC = DIF - DEA, \end{aligned} \tag{7}$$

where $EMA(DIF, signal_period)$ indicates the signal-period smooth moving average of DIF.

By observing the intersection of the MACD line and the signal line, a predictive signal of the market trend can be obtained. When the MACD line crosses the signal line from the bottom up, it is considered a buy signal, indicating that the market may be bullish; Conversely, when the MACD line crosses the signal line from the top down, it is seen as a sell signal, indicating that the market may be bearish. In addition, by looking at the value of the MACD histogram, you can also get a buy or sell judgment. When the MACD histogram is positive, investors get a bullish signal; When the MACD histogram is negative, it often indicates that the stock price has started to fall. It is worth noting that when the MACD line is above the zero line, the price is usually in an upward phase, and when the MACD line is below the zero line, it is generally indicated that the stock price decline has begun.

2.4 XGBoost regression model predicts stock price

XGBoost model, the full name of which is extreme gradient boosting, was first proposed by Dr. Chen Tianqi in 2014, which adds conventional terms on the basis of the original construction, thus improving the calculation accuracy and efficiency of its model to a great extent, and is widely used in predicting stock prices. By using historical stock price data as input, XGBoost automatically finds patterns and trends in price movements during training and makes predictions about future prices accordingly. This forecasting method plays an important role in the financial field. Using XGBoost regression models can help investors and traders better understand the complexity of the stock market and make accurate predictions. By analyzing and learning the patterns of historical stock prices, the model can capture the key features and laws implied in the data, and thus predict the future trend of stock prices. This can provide valuable reference and decision-making basis for investors to help them develop more effective investment strategies and risk management programs. XGBoost's multiple objective functions are as follows:

$$L(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k), \tag{8}$$

where i represents the i th sample in the sample, $l(\hat{y}_i, y_i)$ represents the prediction error of sample i, $\sum_k \Omega(f_k)$ represents the complexity of the tree, i.e. the regular term, which is defined as follows:

$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \|\omega\|^2, \tag{9}$$

where T represents the number of leaf nodes and ω represents the value of the node.

The integration tree model in formula (8) contains functions as parameters, so minimization targets need to be added in Euclidean space, as shown in formula (10):

$$L^{(t)} = \sum_{i=1}^{n} l(y_i, \hat{y}_i^{t-1} + f_t(x_i)) + \Omega(f_t). \tag{10}$$

In order to optimize the objective function quickly, the above expression can be approximated by a second-order Taylor expansion:

$$\tilde{L}^{(t)} \cong \sum_{i=1}^{n} l(y_i, \hat{y}_i^{t-1} + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)) + \Omega(f_t), \tag{11}$$

where $g_i = \partial_{\hat{y}^{(t-1)}} l(y_i, \hat{y}_i^{t-1})$ and $h_i = \partial^2_{\hat{y}^{(t-1)}} l(y_i, \hat{y}_i^{t-1})$ represent the first and second-order gradient statistics of the loss function. By eliminating the constant term within, the above equation can be streamlined to the following form:

$$\tilde{L}^{(t)} \cong \sum_{i=1}^{n} [g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t). \tag{12}$$

Let the set of instances be denoted as $I_i = \{i \mid q(x_i) = j\}$, then incorporating the regularization term formula (9) into equation (12), yields equation (13):

$$\begin{aligned} \tilde{L}^{(t)} &= \sum_{i=1}^{n} l(g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)) + \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{t} \omega_j^2 \\ &= \sum_{j=1}^{T} [(\sum_{i \in I_j} g_i)\omega_j + \frac{1}{2}(\sum_{i \in I_j} h_i + \lambda)\omega_j^2] + \gamma T. \end{aligned} \tag{13}$$

Let $G_j = \sum_{i \in Ij} g_j$ and $H_j = \sum_{i \in Ij} h_j$, then equation (13) can be reformulated into equation (14):

$$\tilde{L}^{(t)} = \sum_{i=1}^{n} [g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{t} \omega_j^2. \tag{14}$$

Subsequently, differentiating Equation (14) with respect to W_j and setting the derivative equal to 0 yields the optimal solution w^* for Equation (14), as illustrated in Equation (15):

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$$w_j^* = -\frac{G_j}{H_j + \lambda}. \quad (15)$$

Upon substituting the optimal solution w_j^* into Equation (14), the resultant objective function is derived, as depicted in Equation (16).

$$\tilde{L}^{(t)} = -\frac{1}{2}\sum_{j=1}^{T}\frac{G_j^2}{H_j + \lambda} + \lambda T. \quad (16)$$

3 Data source

The time series data used in this paper comes from baostock, a third-party data provider that provides Chinese stock market data. Using the API provided by baostock, we log in to the system and inquire the historical K-line data of Shenzhen Stock Exchange stock code "sz.000063". We specify that the start date of query is January 1, 2015, the end date is December 31, 2022, the frequency is daily line (" d "), and the adjustment flag is 3. The result set is iterated over, each record is combined, and the result set is finally output to a CSV file. Based on the data set of ZTE's closing price in the trading days from 2015 to 2022, this paper uses XGBoost model to forecast the stock price and compares it with the real value.

4 Analysis and Prediction

4.1 Analysis of EMA and SMA

Reading the CSV file containing the stock data, calculating the 9-day exponential moving average EM, calculating the 5-day, 10-day, 15-day, and 30-day SMA, Plot these moving averages against the closing price curve using Plotly, as shown in Figure 1 below.

Based on the data shown in Figure 1, we can observe the following. The first is the phenomenon of golden cross and death cross: when the short-term EMA (e.g. 9 days) of ZTE stock crosses from below the long-term SMA (e.g. 30 days), forming a golden cross, it often indicates that the trend of the stock is upward. For example, on dates such as March 2 and March 23, 2018, and June 17 and December 9, 2019, there was a golden cross, which is often seen as a buy signal. When the short-term EMA crosses the long-term SMA from above, forming a death cross, it may mean that the trend of ZTE's stock is heading downward. June 17, 2015, May 15, 2019, October 18, 2019, and December 8, 2020, for example, all saw death crosses, often seen as sell signals.

Secondly, we can judge the trend of the stock by whether the stock price is above or below the EMA: when the ZTE stock price is above the EMA, it usually means that the stock is in an upward trend. For example, during the periods from October 8 to November 5, 2019, October 9 to October 22, 2020, and May 25 to June 29, 2022, the share price was consistently above the EMA. Conversely, when the stock price is below the EMA, it usually indicates that the stock is in a downtrend. For example, between October 18 and October 27, 2017, January 24 to February 18, 2022, and December 12 to December 29, 2022, the stock price was below the EMA.

In addition, we can also observe the slope of the short-term EMA and its distance from the long-term SMA, which reflects the strength of the trend to some extent: when the short-term EMA slope of ZTE stock rises and the distance from the long-term SMA is further, it usually means that the trend of the stock is stronger. For example, during the periods from October 26 to November 24, 2017, February 7 to March 13, 2020, and July 8 to July 21, 2021, stocks showed a stronger trend. Conversely, when the short-term EMA slope is declining and the distance to the long-term SMA is close, it may indicate a weaker trend in the stock. For example, during the periods from July 11 to August 1, 2018, April 10 to April 29, 2020, and August 26 to September 16, 2021, stocks exhibited a weaker trend.

To sum up, based on the exponential moving averages (EMA) and the simple moving average (SMA) analytical expression templates can be by looking at the golden cross and death cross located above or below the EMA, stock prices and the slope of short-term EMA and judging and long-term factors such as the distance of the SMA zte stock trend strength and buy and sell signals.

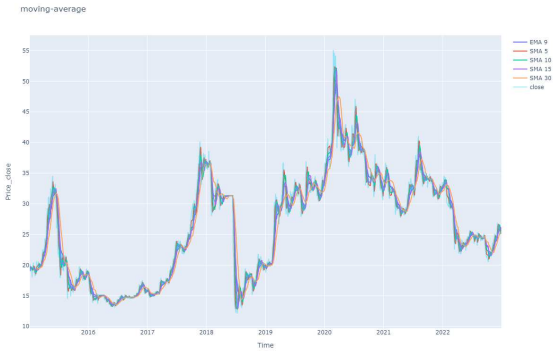


Fig. 1. EMA, SMA and closing price time series

4.2 Analyzing RSI through visual chart

RSI is a measure of the strength of price movements and is often used to analyze overbought and oversold situations in stocks and to aid in the development of trading strategies. By calculating RSI indicators and drawing charts for detailed analysis, as shown in Figure 2 below, we can draw the following conclusions. First, based on the value of the RSI indicator, we can identify overbought and oversold areas. When the RSI index exceeds 80, it enters the overbought area, which may indicate that ZTE shares have been overbought and there may be opportunities for a correction or correction. For example, on or about March 21, 2016, on or about May 4, 2018, and February 24, 2020. Conversely, when the RSI indicator is below 20, it enters oversold territory, which may indicate that ZTE shares have been over-sold and there may be opportunities for a rebound or reversal. For example, on or about April 3, 2016, on or about July 8, 2018, and on or about March 13, 2020.

And then, we can judge the strength of the stock trend based on the movement of the RSI indicator. When the RSI index of ZTE's stock continues to rise and stabilize above 50, it may indicate a stronger stock trend. For example, around March 2015, around February 2019, and around July 2021. On the contrary, when the RSI indicator of ZTE's stock continues to decline and stabilize below 50, it may indicate a

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weaker stock trend. For example, around November 2018, around May 2019, and around August 2020.

Finally, we can also observe the reversal signal of the RSI indicator to determine the timing of a possible reversal or correction of the stock. When the RSI indicator of ZTE stock falls within the overbought area and breaks through the key level, it may indicate that there may be an opportunity for a reversal or correction in the stock. For example, April 7, 2016 and May 7, 2018. On the contrary, when the RSI indicator of ZTE stock rises within the oversold area and breaks through the key level, it may indicate that the stock may have an opportunity for a rebound or reversal. For example, March 6, 2016 and June 12, 2018.

In summary, by analyzing the RSI indicators of ZTE shares, we can draw conclusions about overbought and oversold areas, trend strength, and reversal signals, providing reference for investors to develop trading strategies.

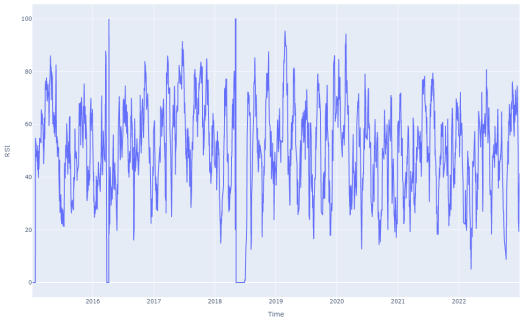


Fig. 2. Time series diagram of RSI

4.3 Plotting MACD and conducting analysis

MACD is a commonly used technical indicator to measure changes in price momentum and trend. By plotting the MACD indicator and its signal line in the chart, you can visually observe the change of MACD. In this paper, the 12-day exponentially smoothed moving average (EMA_12) and 24-day exponentially smoothed moving average (EMA_24) of the closing price are calculated, and the difference between EMA_12 and EMA_24 is the MACD and its Signal line, as shown in Figure 3 below. The histogram of MACD is made with the difference between MACD and signal line, as shown in Figure 4 below.

The rise and fall of the MACD line can be used to determine the trend of the stock price. When the MACD line is above the zero line and rising, it indicates that the upward trend of the stock price may be more obvious, such as around November 2017, around February 15, 2019, and February and March 2020. When the MACD line is below the zero line and falling, it indicates that the downward trend of the stock price may be more obvious, such as around July 2015, and around June 2018. The crossing of the MACD line with the signal line can be used to determine the timing of turning points or trend reversals in the stock price, providing buy and sell signals. When the MACD line crosses the signal line upward, a buy signal is generated, such as October 25, 2017, March 19, 2019; When the MACD line crosses down the signal line, a sell signal is generated, such as September 24, 2019, and July 9, 2020.

The MACD Histogram can be used to measure the momentum of a stock's price. The height of the bar chart represents the difference between the MACD line and the signal line. When the bar chart becomes larger, it indicates that the price momentum has increased, and there may be more obvious fluctuations, such as July 2015, around June 15, 2018, and around March 2020. When the bar becomes smaller or crosses the zero line, it indicates that price momentum has weakened and may enter a sideways or volatile market, such as around early September 2016, around mid-January 2017, and March 2021.



Fig. 3. MACD, Signal line and closing price time series chart

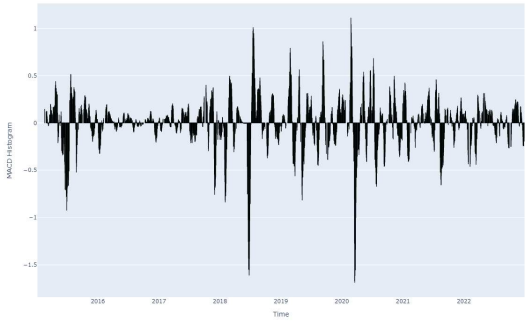


Fig. 4. Histogram of MACD

4.4 XGBoost regression model for stock price prediction

XGBoost is a popular machine learning algorithm that has certain advantages in predicting stock prices. Based on the idea of gradient tree, XGBoost can deal with complex nonlinear relations, and can automatically select features and deal with missing values. Therefore, this paper uses XGBoost algorithm to predict stock prices.

First, we divide the data set into training set and validation set according to the proportion of 70% and 30%, define the grid parameter dictionary, including the hyperparameters of the model, which are respectively 'gamma', 'learning_rate', 'max_depth', 'n_estimators', 'random_state', GridSearchCV is used to perform parameter search. GridSearchCV automatically traverses these parameter combinations and returns the best parameter combination and the corresponding model. The best parameter combination is 'gamma'=0.01, 'learning_rate'=0.05. 'max_depth'=15, 'n_estimators'=300, 'random_state'=42. GridSearchCV performs cross-validation to evaluate the performance of each parameter combination. After fitting the model on the training set, it calculates the performance score of the model

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on the verification set by calling the score function, and the calculated score is 0.6718, indicating that this model has certain good performance in predicting the closing price of stocks. In order to clearly show the influence of the nine features Close, EMA_9, SMA_5, SMA_10, SMA_15, MSA_30, RSI, MACD and MACD_signal on the prediction of stock prices, we drew the importance ranking diagram of each feature in the XGBoost model, as shown in Figure 5 below.

Finally, we initialize a new XGBoost regression model by obtaining the best combination of parameters. The model is trained with the best parameters and makes predictions on the validation set. The prediction results are shown in Figure 6 below. To evaluate the performance and stability of the model, we calculated the mean absolute error MAE=0.6416, coefficient of determination R2=0.9667, root mean square error RMSE=0.0278, and root mean square logarithm error RMSLE=0.0278. The calculated results have small mean absolute error and root-mean-square error, high coefficient of determination and small root-mean-logarithm error. The results of these indexes show that the predicted results of the model are close to the actual values, and the model has high explanatory ability and stability.

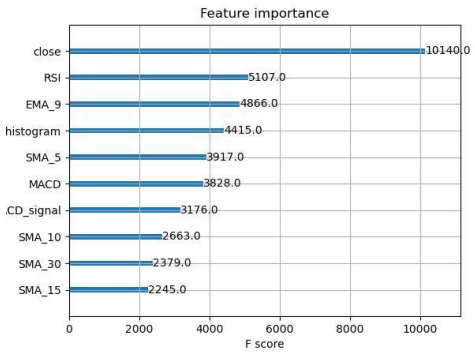


Fig. 5. Feature importance ranking



Fig. 6. A time series plot of the true and predicted closing values of a stock

5 Conclusion

Based on the trading data set between 2015 and 2022, this study has used the XGBoost model to make a forecast of ZTE's stock price and compared it with the actual value. The research has mainly included the following aspects: First, the EMA and SMA were calculated and presented, which were helpful to identify the long-term trend and short-term volatility of stock prices. Second, it analyzed the RSI, which measured overbought and oversold conditions in stock prices to determine whether prices have risen or fallen excessively. Next, the MACD was calculated and displayed, which contributed to identify trends in stock prices and smooth out short-term fluctuations. Finally, XGBoost regression model has been used to predict the stock price. XGBoost was a powerful machine learning algorithm that could learn patterns and trends in historical data, predict future stock price changes, and calculate relevant evaluation indicators. The results have shown that the predicted results of the model were relatively close to the actual value, and the model has high explanatory power and stability. This paper has provided a stock price prediction method based on XGBoost model, and verified its effectiveness in practical application, which has important reference value for investors and financial institutions in formulating investment strategies and managing risks.

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